

# KUSHAGRA SHARMA

## Software Engineer | Python Backend | Cloud APIs | GenAI/RAG

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### SUMMARY

Associate Software Engineer at E2E Networks (E2E Cloud) working across Python backend engineering, cloud platform APIs, AI-native documentation, billing workflow architecture, API automation, and LLM-assisted internal tooling. Experienced with Django/FastAPI services, REST APIs, OpenAPI tooling, Pytest automation, GenAI/RAG applications, Docker, Kubernetes, AWS/Azure exposure, MySQL, MongoDB, and Qdrant. Seeking Software Engineer, Python Backend Engineer, AI Engineer, GenAI/RAG Engineer, and Cloud Platform Engineer roles.

### TECHNICAL SKILLS

- Backend and APIs: Python, Django, FastAPI, Flask, REST APIs, Django REST Framework, OpenAPI/Swagger, Pytest, Allure, Streamlit
- AI and LLM: LLMs, RAG, LangChain, Prompt Engineering, Qdrant, Sentence Transformers, Groq Llama, AutoGen, model evaluation
- Cloud and DevOps: E2E Cloud, AWS, Azure, Docker, Kubernetes, CI/CD, Git, Linux, Nginx
- Data and ML: MySQL, PostgreSQL, MongoDB, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch, OpenCV

### PROFESSIONAL EXPERIENCE

#### E2E Networks (E2E Cloud) - NSE-listed cloud and GPU infrastructure provider | Delhi, India

##### Associate Software Engineer | Apr 2026 - Present

- Helped establish E2E's AI-native documentation layer by restructuring cloud service docs, API references, and developer guides into retrieval-friendly content for stronger LLM discoverability, chatbot/agent accessibility, and Ask AI support.
- Built an in-repo API documentation pipeline with DSL/OpenAPI compilation, SSR documentation routes, generated code samples, schema views, interactive Try It workflows, and AI-consumable indexes for cloud platform services.
- Working on architectural improvements to long-running billing workflows, focusing on execution-time reduction, scalability, maintainability, and operational reliability across usage, pricing, and billing systems.
- Re-architected the Parallel File System (EPFS) Django service onto a cluster-driven model, redesigning core schemas, delivering 8+ production migrations with backfills, and improving billing accuracy through SKU price snapshotting.
- Refactored 9 duplicated AWX orchestration job classes into a shared parameterized framework, reducing maintenance overhead across infrastructure provisioning and billing workflows.
- Expanded Python/Pytest API automation across VPC, Kubernetes, compute, file systems, container registry, and GenAI workflows to improve API validation consistency and regression visibility.

##### SDET - AI & API Automation | Jul 2025 - Apr 2026

- Architected Python/Pytest/Allure API automation for VPCs, Kubernetes, file systems, compute, container registry, and GenAI fine-tuning workflows, reducing manual validation effort by about 70%.
- Built LLM-assisted test analysis tooling for execution summaries, failure-context analysis, and API-change-based suite recommendations through a Streamlit orchestration UI backed by object storage.
- Improved API reliability and release confidence through OpenAPI-based validation, REST API coverage, failure analysis, and structured release notes for cloud platform services.
- Collaborated with backend, DevOps, and product teams on cloud API documentation, developer workflows, internal tooling, bug triage, and release validation.

##### SDET Intern | Jan 2025 - Jul 2025

- Built and maintained API automation for infrastructure services including VPCs, Kubernetes, shared file systems, compute nodes, container registry, and GenAI fine-tuning workflows.
- Created scalable test structures, review checklists, and release validation notes to improve automation maintainability and team execution quality.
- Validated cloud-native platform features such as node scaling, VM lifecycle operations, file-system quotas, registry access controls, and orchestration workflows.

##### WhatBytes | Engineering Intern - Backend | Nov 2024 - Jan 2025 | Remote

- Built Django and PostgreSQL backend workflows for Thirdi.ai, covering campaign creation, campaign management, authentication, and REST API integrations.
- Integrated TikTok Ads, Google Ads, LinkedIn, and Stripe for ad operations, reporting, payments, and secure platform access.
- Developed real-time metric pipelines and reporting tools that helped non-technical users inspect campaign performance and operational status.

##### StoryVord Ltd. | AI & Backend Intern | May 2024 - Jul 2024 | Remote

- Built 3 of 5 LLM-powered assistant bots for film-production workflows: compliance guidance, local culture guidance, and logistics planning.
- Developed Django REST APIs, authentication flows, and bot-to-service orchestration for Azure-hosted backend systems.
- Collaborated across AI, frontend, and backend teams while contributing to architecture and scalability improvements.

### PROJECTS

#### RAG Patent Q&A System | FastAPI, Qdrant, LangChain, Groq Llama, Docker, AWS

- Built a domain-specific RAG assistant over Indian patent law documents with PDF ingestion, recursive chunking, bge-large embeddings, Qdrant vector search, FastAPI interaction layer, and source-attributed answers.
- Indexed 1,000+ document chunks and deployed the system with Docker and environment-managed credentials.
- Live demo: <https://huggingface.co/spaces/kushagrasharma-13/patent>

#### AI-Powered Voice Assistant | Python, Flask, MongoDB, Azure Cognitive Services, AutoGen

- Led a 5-member team in the Code Crunch 2024 National Hackathon to build a modular voice/text assistant for app control, messaging, email generation, reminders, and real-time information retrieval.

- Designed a plugin-style flow covering speech-to-text, natural language understanding, skill execution, response generation, text-to-speech, and MongoDB-backed sessions.

#### **StoryVord AI Assistants | Django REST, LLMs, Azure**

- Built backend APIs and assistant logic for compliance, culture, and logistics bots used to support film-production planning.
- Integrated LLM-assisted guidance with backend services and frontend workflows across an Azure-hosted application.

#### **Prakriti - Plant Disease Detection and Remedy System | PyTorch, Flask, AWS EC2**

- Led backend development for an 8-hour hackathon project that detects plant diseases from leaf images and returns diagnosis, prevention steps, and treatment suggestions.
- Built the model-serving API with PyTorch and Flask and deployed the stack on AWS EC2.

### **EDUCATION**

GLA University, Mathura | B.Tech Computer Science, Artificial Intelligence and Machine Learning | 2021 - 2025 | CGPA: 8.66/10

### **RECOGNITION AND CERTIFICATIONS**

- First Prize - Smart India Hackathon Senior Software Edition
- IBM National Hackathon Finalist 2024
- Top 1% - Algorithms and Analysis, IIT Kharagpur
- Certifications: DevOps Tools Using AWS, NPTEL Design and Analysis of Algorithms, Advanced NLP with Python, Career Essentials in Generative AI by Microsoft